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标题（中文）: 最小作用量原理的拓扑推导
Subtitle（英文））: First-Principles Proof from the Perspective of Angular-Momentum Network Geometry (ANG)
副标题（中文）: 角动量网络几何学（ANG）视角下的第一性原理证明

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Abstract 摘要

English
The principle of least action is one of the deepest principles in physics, yet conventionally it is treated as an axiom — incapable of being derived from something more fundamental. Within the Angular-Momentum Network Geometry (ANG) framework, this paper demonstrates that the principle of least action is not a "preference" of nature, but a rigid corollary of Axiom 0 (Global Zero-Sum Angular Momentum) at the level of link networks.

We prove rigorously:

1. The action $\mathcal{S}$ equals the total curvature of the link network on the 6-dimensional angular-momentum manifold;
2. Any non-extremal path generates an uncanceled angular-momentum residual $\Delta\mathbf{J} \neq 0$;
3. Axiom 0 rigidly forbids any $\Delta\mathbf{J} \neq 0$;
4. Consequently, the link network must select paths that extremize $\mathcal{S}$.

Keywords: principle of least action; topological derivation; Angular-Momentum Network Geometry; global zero-sum angular momentum; 6D angular-momentum manifold

中文
最小作用量原理是物理学中最深层的原理之一，但传统上它被视为 "公理"—— 无法从更基础的东西推导出来。本文在角动量网络几何学（ANG）框架中证明：最小作用量原理不是自然界的 "偏好"，而是 Axiom 0（全局角动量归零）在链接网络层面的刚性推论。

我们严格证明：

1. 作用量 $\mathcal{S}$ = 链接网络在 6D 角动量流形上的总曲率；
2. 任何非极值路径都会产生未被抵消的角动量残差 $\Delta\mathbf{J} \neq 0$；
3. Axiom 0 刚性禁止任何 $\Delta\mathbf{J} \neq 0$；
4. 因此，链接网络必须选择使 $\mathcal{S}$ 取极值的路径。

关键词：最小作用量原理；拓扑推导；角动量网络几何学；全局角动量归零；6D 角动量流形

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1. Introduction 引言：最小作用量原理的未解之谜

English
The principle of least action stands as the crown jewel of physics:

$$\delta\mathcal{S} = 0, \mathcal{S} = \int_{t_1}^{t_2} L, dt$$

It unifies Newtonian mechanics, electromagnetism, relativity and quantum mechanics — nearly all of physics is built upon it. Nevertheless, physicists have never explained why nature must obey this principle.

Feynman's path-integral formulation re-interprets "action extremization" as superposition of phase-coherent paths, yet this only transforms the question from "why extremal action?" into "why phase coherence?"; the fundamental origin remains unanswered.

This paper answers that question: the principle of least action is not an axiom, it is a necessary consequence of Axiom 0.

中文
最小作用量原理（Principle of Least Action）是物理学的皇冠：

$$\delta\mathcal{S} = 0, \mathcal{S} = \int_{t_1}^{t_2} L, dt$$

它统一了牛顿力学、电磁学、相对论、量子力学 —— 几乎整个物理学都建立在它之上。然而，物理学家从未解释过为什么自然必须遵循这个原理。

费曼的路径积分将 "作用量极值" 转化为 "相位相干路径的叠加"，但那只是将问题从 "为什么作用量极值？" 转化为 "为什么相位相干？"—— 并没有回答根本问题。

本文回答这个问题：最小作用量原理不是公理，它是 Axiom 0 的必然结果。

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2. The Action within the ANG Framework ANG 中的作用量

2.1 Action = Total Curvature of the Link Network 作用量 = 链接网络的总曲率

English

In standard physics the action $\mathcal{S} = \int L, dt$ is an abstract quantity. Within ANG, it is reduced to the total topological curvature of the link network over the 6-dimensional angular-momentum manifold:

$$\boxed{\mathcal{S} = \int_{\text{path}} \mathbf{J} \cdot d\mathbf{x}}$$

Where:

- $\mathbf{J} = \mathbf{L} + \mathbf{S}$: local angular-momentum flux carried by the link;
- $d\mathbf{x}$: displacement of the link on the 6D manifold.

A direct corollary of this formula is that every configuration of the link network corresponds to a well-defined total curvature $\mathcal{S}$.

中文

在标准物理学中，作用量 $\mathcal{S} = \int L, dt$ 是一个抽象量。在 ANG 中，它被还原为链接网络在 6D 角动量流形上的总拓扑曲率：

$$\boxed{\mathcal{S} = \int_{\text{路径}} \mathbf{J} \cdot d\mathbf{x}}$$

其中：

- $\mathbf{J} = \mathbf{L} + \mathbf{S}$：链接携带的局域角动量通量；
- $d\mathbf{x}$：链接在 6D 流形上的位移。

这个公式的直接推论是：任何链接网络的构型都有一个对应的总曲率 $\mathcal{S}$。

2.2 Variation of Action = Residual Angular-Momentum 作用量的变分 = 角动量的残差

English

If the link network shifts from path P_1 to an alternative path P_2, the change in action equals the difference in angular-momentum flux between the two paths:

$$\delta\mathcal{S} = \int_{P_1}^{P_2} \mathbf{J} \cdot d\mathbf{x} = \Delta\mathbf{J}_{\text{path}}$$

In other words: the variation of the action $(\delta\mathcal{S})$ is exactly the residual angular momentum $(\Delta\mathbf{J})$ between competing paths.

中文

如果链接网络从一条路径 P_1 变化到另一条路径 P_2，作用量的变化等于两条路径之间角动量通量的差：

$$\delta\mathcal{S} = \int_{P_1}^{P_2} \mathbf{J} \cdot d\mathbf{x} = \Delta\mathbf{J}_{\text{路径}}$$

也就是说：作用量的变分（$\delta\mathcal{S}$）就是路径之间的角动量残差（$\Delta\mathbf{J}$）。

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3. Rigid Constraint from Axiom 0 Axiom 0 的刚性约束

3.1 Restatement of Axiom 0 Axiom 0 的重新表述

English

Axiom 0 (Global Zero-Sum Angular Momentum) requires:

$\mathbf{J}_{\text{total}} \equiv 0$

This means any local angular momentum must be completely canceled at the global level. There is no allowance for “approximate cancellation” or “mostly canceled”.

中文

Axiom 0（全局角动量归零）要求：

$\mathbf{J}_{\text{total}} \equiv 0$

这意味着：任何局域角动量必须在全局被完全对消。不存在 “近似归零” 或 “大部分归零” 的余地。

3.2 Physical Consequence of Non-Extremal Paths 非极值路径的物理后果

English

Suppose the link network follows a non-extremal path ($\delta\mathcal{S} \neq 0$):

1. A non-extremal path implies a non-zero angular-momentum difference $\Delta\mathbf{J} \neq 0$ between paths;
2. $\Delta\mathbf{J} \neq 0$ means angular momentum along this path is not fully canceled;
3. Uncanceled angular momentum constitutes a “residual”;
4. Axiom 0 rigidly forbids any residual;
5. Therefore non-extremal paths are inadmissible.

中文

假设链接网络选择一条非极值路径（即 $\delta\mathcal{S} \neq 0$）：

1. 非极值路径意味着路径之间存在非零的角动量差 $\Delta\mathbf{J} \neq 0$；
2. $\Delta\mathbf{J} \neq 0$ 意味着这条路径上的角动量没有被完全对消；
3. 没有被完全对消的角动量是 “残差”；
4. Axiom 0 刚性禁止任何残差；

5. 因此，非极值路径是不允许的。

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4. Derivation: From Axiom 0 to the Principle of Least Action 推导：从 Axiom 0 到最小作用量原理

English

Step	Logic	Result
1	Assume the link network evolves along path P	—
2	Path P possesses total curvature $\mathcal{S}_P = \int_P \mathbf{J} \cdot d\mathbf{x}$	—
3	Consider a neighboring alternative path P'	—
4	Curvature difference between the two paths: $\delta\mathcal{S} = \mathcal{S}_{P'} - \mathcal{S}_P$	—
5	$\delta\mathcal{S} \neq 0$ is equivalent to $\Delta\mathbf{J} \neq 0$	Angular-momentum residual exists
6	Axiom 0 forbids any $\Delta\mathbf{J} \neq 0$	Rigid prohibition
7	Thus path P must satisfy $\delta\mathcal{S} = 0$	Extremal path
8	$\delta\mathcal{S} = 0$ is exactly the principle of least action	Derivation complete

$\boxed{\text{Axiom } 0 \Longrightarrow \delta\mathcal{S} = 0}$

中文

步骤	逻辑	结果
1	假设链接网络沿路径 P 演化	—
2	路径 P 有总曲率 $\mathcal{S}_P = \int_P \mathbf{J} \cdot d\mathbf{x}$	—
3	考虑另一条相邻路径 P'	—

4	两条路径的曲率差 $\delta\mathcal{S} = \mathcal{S}_{P'} - \mathcal{S}_P$	—
5	$\delta\mathcal{S} \neq 0$ 等价于 $\Delta\mathbf{J} \neq 0$	存在角动量残差
6	Axiom 0 禁止任何 $\Delta\mathbf{J} \neq 0$	刚性禁令
7	因此，路径 P 必须使 $\delta\mathcal{S} = 0$	极值路径
8	$\delta\mathcal{S} = 0$ 就是最小作用量原理	推导完成

$\boxed{\text{Axiom } 0 \Longrightarrow \delta\mathcal{S} = 0}$

5. Reformulation of the Physical Meaning 物理含义的重新表述

5.1 Standard Statement versus ANG Statement 标准表述 vs ANG 表述

English

Standard Physics	ANG (Geometric Ontology)
“Nature chooses the path that minimizes the action”	“Nature forbids angular-momentum residuals”
Least action is a “preference”	Least action is a “prohibition”
Why? Unknown	Why? Because of Axiom 0

中文

标准物理	ANG（几何本体）
“自然选择使作用量最小的路径”	“自然不允许角动量残差”
最小作用量是 “偏好”	最小作用量是 “禁令”
为什么？不知道	为什么？因为 Axiom 0

5.2 The Deep Picture of Physics 物理学的深层图景

English
Within ANG, cosmic evolution is not “particles moving along optimal trajectories”. Instead:

> The link network continuously adjusts its configuration to eliminate all angular-momentum residuals in order to maintain global zero-sum angular momentum. The macroscopic projection of this adjustment process is “objects moving along geodesics” — the principle of least action.

中文
在 ANG 中，宇宙的演化不是 “粒子沿着最优路径运动”，而是：

> 链接网络为了维持全局角动量归零，不断调整自身构型以消除所有角动量残差。这个调整过程在宏观上的投影就是 “物体沿测地线运动”—— 即最小作用量原理。

5.3 Natural Connection to Quantum Mechanics 与量子力学的自然衔接

English
In the path-integral formulation of quantum mechanics, the phase factor for every path is $e^{i\mathcal{S}/\hbar}$. When $\mathcal{S}$ departs far from its extremum, phases oscillate rapidly and cancel out; only paths near $\delta\mathcal{S} = 0$ constructively interfere.

Within ANG this phenomenon is reduced: paths far from the extremum carry uncanceled angular-momentum residuals and are therefore “filtered out” during global cancellation. The “phase coherence” of quantum mechanics is the probabilistic projection of Axiom 0.

中文
在路径积分（量子力学）中，所有路径的相位是 $e^{i\mathcal{S}/\hbar}$。当 $\mathcal{S}$ 远离极值时，相位快速振荡，相干相消。只有 $\delta\mathcal{S} = 0$ 附近的路径相干相长。

在 ANG 中，这个现象被还原为：远离极值的路径携带未被抵消的角动量残差，因此在全局对消过程中被 “滤除”。量子力学的 “相位相干” 本质上是 Axiom 0 在概率层面上的投影。

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6. Falsifiable Predictions 可证伪预测

English

1. If a physical system is found to evolve along a trajectory violating the principle of least action, then Axiom 0 of ANG is invalidated.

2. If experiments discover that angular-momentum residuals can remain permanently localized without global cancellation (permanent local angular momentum), Axiom 0 fails and the ANG framework is falsified.

3. If non-extremal paths within the quantum-mechanical path integral produce macroscopically observable effects incompatible with the constraints of Axiom 0, the ANG framework requires revision.

中文

1. 如果存在一个物理系统，其演化路径不满足最小作用量原理，则 ANG 的 Axiom 0 失效。

2. 如果实验发现角动量残差可以不被全局对消（即存在 “永久局域角动量”），则 Axiom 0 失效，ANG 框架被证伪。

3. 如果量子力学中的非极值路径路径积分贡献可以产生宏观可观测效应，且该效应与 Axiom 0 的约束不兼容，则 ANG 需要修正。

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7. Conclusions 结论

English

Within the Angular-Momentum Network Geometry (ANG) framework, we have rigorously proven:

The principle of least action is not an axiom of nature, but a necessary corollary of Axiom 0 (Global Zero-Sum Angular Momentum) at the level of link networks.

The complete logical chain of the proof:

$$\boxed{\text{Axiom 0} \xrightarrow{\mathcal{S}=\int \mathbf{J}\cdot dx} \delta\mathcal{S} = \Delta\mathbf{J} \xrightarrow{\text{Axiom 0 forbids}\Delta\mathbf{J}\neq 0} \delta\mathcal{S} = 0}$$

The physical principle of least action may be restated in ANG language:

To maintain global zero-sum angular momentum, the link network must follow paths with minimal total curvature. Any deviation generates residual angular momentum, which cannot exist within the 6-dimensional angular-momentum manifold because it is forbidden by Axiom 0.

This is the true origin of the principle of least action — it is not a physical “principle”, but a geometric “enforcement”.

中文

在角动量网络几何学（ANG）框架中，我们严格证明了:

最小作用量原理不是自然界的公理，而是 Axiom 0（全局角动量归零）在链接网络层面的必然推论。

证明的完整逻辑链:

$$\boxed{\text{Axiom 0} \xrightarrow{\mathcal{S}=\int \mathbf{J}\cdot d\mathbf{x}} \delta\mathcal{S} = \Delta\mathbf{J} \xrightarrow{\text{Axiom 0 禁止 } \Delta\mathbf{J}\neq 0} \delta\mathcal{S} = 0}$$

物理学的“最小作用量原理”在 ANG 中可以被重新表述为：

链接网络为了维持全局角动量归零，必须选择总曲率最小的路径。任何偏离都会产生残余角动量，而残余角动量在 6D 角动量流形中无法存在，因为 Axiom 0 禁止它。

这就是最小作用量原理的真正原因——它不是一个物理“原理”，而是一个几何“强制”。

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